Final Project Step 2

Caitie Schrotberger

2024-05-27

Introduction

Cornoavirus (C19) hit the world hard in the beginning of 2020. This highly contagious virus started in China and quickly spread across the world. As the disease progressed it mutated into many different variants making it much harder to treat. It quickly became a public health emergency and classified as a pandemic. Pandemics only hit the world about once every 100 years. Cornavirus left the world struggling to react. It didn't know how to treat the disease, which demographics of people should be the most at risk.

As Public Health professionals across the globe responded to this new virus, they had no prior information to go off of. During the last pandemic, the world was in a much different technological and communicative level. The Spanish Flu of 1918 occurred in a time with limited medical care, no method of fast communication across the world, and no technology which made it very difficult to track the true impact to the world, especially in death counts.

Now that the pandemic is over, while the virus is still spreading and mutating, because of the way the world responded to the pandemic we now have much better information and data to be able to analyze how specific response methods impacted the spread of the disease and which deomgraphics (age, gender, race, ethnicity, location) were more succeptible to contracting it. To be able to plan for and potentially have a more favorable outcome to the next pandemic, we can use data from this incident to create models that could be re-used during the next pandemic to react accordingly.

Problem Statement

The problem that this research will address is how to identify high risk people during a pandemic.

By not preparing a way to identify high risk populations during a pandemic, the United States opens itself up to high hospitalizations, stressing the capacity and staffing of facilities. Additionally high death rates which would strain morgues and funeral homes would put additional stress on a population already under tension by losing loved ones.

Data and research from prior pandemics will allow creation of predictive regression models that can be used for emergency response. Having models pre-built will allow for use and potential model re-training to make it more accurate. This will allow response theams to implement control measures more quickly and effectively on high risk populations.

Research questions

Here are potential research questions that focus on the problem topic:

- How fast did C19 spread in the USA?
- Were there demographic groups that were more likely to contract C19?
- How many people were hospitalized from C19 and what were the demographics of them?
- Did certain populations have a higher propensity for death from C19?
- Did the vaccine help control the spread of the disease?
- Were certain demographic groups more likely to benefit from the vaccination?
- Did the governmental mandates of large scale quarantines help slow the spread of C19?

Analysis

Approach

To be able to appropriately identify those at highest risk of contracting, being hospitalized, or dying from a future pandemic outbreak you need to model based on variables that will be available at any point in the future. Using consistent demographic variables (such as age, gender, race, ethnicity, and geolocation) along with the values looking to be predicted (such as contract, hospitalized, and deceased) are factors that are currently standard health care metrics that can be easily obtained.

Separate models should be built for each health outcome independently and then together. Testing the simple and multiple regression models against each other will help us understand how to better help people. For example creating a model to predict those that might die from a virus out of the entire population gives you a small percentage of the targeted variable, but if you start with a data set based on those that were hospitalized, it may provide a greater model as the proportion of the targeted audience to the population is smaller.

How your approach addresses the problem.

The spread of viruses are hard to manage and can be unpredictable. To be able to partially address the spread and impact to society of a new virus having multiple models created to reuse and modify during the identification and spread of a new pandemic will allow for faster public health response.

Data will be obtained from public data sources on the number of people that contracted C19, the number hospitalized, and the number deceased. Demographics such as age, gender, race, ethnicity, and location will also be part of those data sets.

The data will be analyzed via plots and graphs to look for trends and relationships between the variables. From there a final data set with a predictor variable for hospitalization and death will be created.

Different models will be needed because not everyone that passes away makes it to a hospital. So predictions could be for hospitalization, hospitalization and death, and just death.

Data (Minimum of 3 Datasets)

- 1: Census data by state and age and gender
  - https://data.census.gov/table/DECENNIALDP2020.DP1?g=010XX00US$0400000_040XX00US31&d=DEC%20Demographic%20Profile
  - This file is sourced by the US Census Bureau, the 2020 Decennial Census.
  - It provides by State, the population counts broken down by age (5 year buckets), gender, race, ethnicity, household type, and housing type. The counts and the percents (distributions) by category are provided.
- 2: CDC COVID Hospitalizations by Jurisdiction
  - https://data.cdc.gov/Public-Health-Surveillance/United-States-COVID-19-Hospitalization-Metrics-by-/39z2-9zu6/about_data
  - This file is sourced by the Centers for Disease Control and Prevention (CDC). It was sourced by information provided directly from hospitals across the US. Data was collected from 8/1/2020 through 5/3/2024.
  - It provides the date of data collection, HHS Region number, number of new C19 Hospital Admission, Cumulative C19 Hospital Admissions, Total Hospitalized C19 Patients, C19 Inpatient Bed Occupancy (7 day average), C19 ICU Bed Occupancy (7 day average).
- 3: COVID Cumulative Deaths by state and gender and age
  - https://data.cdc.gov/NCHS/Provisional-COVID-19-Deaths-by-Sex-and-Age/9bhg-hcku/about_data
  - This file is sourced by the Centers for Disease Control and Prevention (CDC). It was sourced by information provide by the Nation Center for Health statistics.
  - It provides data by month, year, and total of C19 Deaths, All Deaths, Pneumonia Daths, Pneumonia and C19 Death, Influenza Deaths, Pneumonia/Influenza/C19 Deaths broken down by gender and age.
- 4: COVID cumulative cases by week by state
  - https://data.cdc.gov/Case-Surveillance/Weekly-United-States-COVID-19-Cases-and-Deaths-by-/pwn4-m3yp/about_data
  - This file is sourced by the CDC. It received information directly from the states on a daily basis from August 2020 through April 2024.
  - It provides data by day and HHS region including
- 5: COVID Case level by case, incl month, age, gender, hosp, death
  - https://data.cdc.gov/Case-Surveillance/COVID-19-Case-Surveillance-Public-Use-Data-with-Ge/n8mc-b4w4/about_data
  - This file is sourced by the CDC through information received by the states from January 2020 through May 2023
  - It provides date received, state, range data represents (generally a week), total cases, new cases, total deaths, new deaths all related to C19.
- 6: Vaccination statuses by date and age
  - https://data.cdc.gov/Vaccinations/COVID-19-Vaccinations-in-the-United-States-County/8xkx-amqh/about_data
  - This file is sourced by the C19 Vaccine Administration process and includes data from clinics, pharmacies, long-term care facilities, dialysis centers, Federal Emergency Management Agency (FEMA), and other partner sites.

    - It provides data by State, county, and week for the percent of the population that had received 1 dose or a complete series of doses broken down by large age groups. It also has data on boosters and census population.

Overall these data sets provide the population demographics by state age and gender, then many different data sets on incidence (cases), hospitalization, death, and vaccination efforts.

Required Packages

The R packages needed for this project are as follows

```
library(rmarkdown)
library(readxl)
library(openxlsx)
library(readr)
library(shiny)
library(tinytex)
library(knitr)
library(latexpdf)
library(ggplot2)
library(plyr)
library(dplyr)
```

```
##
## Attaching package: 'dplyr'

## The following objects are masked from 'package:plyr':
##
##     arrange, count, desc, failwith, id, mutate, rename, summarise,
##     summarize

## The following objects are masked from 'package:stats':
##
##     filter, lag

## The following objects are masked from 'package:base':
##
##     intersect, setdiff, setequal, union
```

How to import and clean data

The steps to obtain and clean data are similar to a degree. Data cleansing can vary based on the source, quality, and metrics is the data set. To obtain, prepare, and clean the data the following steps will be used.

- Obtain hyperlink to data or download to local directory
- Open and view the data in excel or notepad to view it to understand the data structure
- Write R code to import the data file into a data frame
- Review each data source by looking at the first records (head)
- Remove duplicate rows

- Correct any spelling errors
- Remove or adjust any outliers
- Reorder columns or remove columns not needed
- Remove rows not needed (in cases where dataset provide more than population looking at)
- Adjust values to categorical rather than sequential as needed.

- Ensure have nation wide totals

What does the final data set look like?

Each dataset has been initially cleaned and the head function run which provides the first 6 rows of data.

```
#Data Set 1:  Census data by state and age and gender
URL1 <- "DECENNIALDP2020DP1.xlsx"
DS1 <- read.xlsx(URL1, sheet=2)
DS1.1 <- select(DS1, -contains("X"))
DS1.2 <- select(DS1.1, -contains("rico"))
DS1.3 <- DS1.2[-c(1:2,22:27,47:52,72:88,92:175), ]
DS1.4 <- DS1.3 %>% replace(is.na(.), 0)
DS1.5 <- DS1.4 %>% mutate_at(c(2:50), as.double)
DS1.6 <- DS1.5 %>% mutate(Total = select(., Alabama:Wyoming) %>% rowSums(na.rm = TRUE))
DS1.7 <- DS1.6 %>% relocate(Total, .before=Alabama)
head(DS1.7)
```

```
##              Label     Total Alabama Alaska Arizona Arkansas California
## 3 Total population 300186254 5024279 733391 7151502  3011524   39538223
## 4    Under 5 years  16466169  286529  48104  392370   179575    2137439
## 5     5 to 9 years  17992893  302637  51054  443878   192794    2393219
## 6   10 to 14 years  19317907  325031  51344  485297   205837    2613891
## 7   15 to 19 years  19735414  338475  47433  485891   204915    2644071
## 8   20 to 24 years  19982555  345931  49456  477713   198109    2731553
##   Colorado Connecticut Delaware District.of.Columbia  Florida  Georgia  Hawaii
## 3  5773714     3605944   989948               689545 21538187 10711908 1455271
## 4   314580      176831    51230                37068  1030284   614218   77352
## 5   347525      197636    56920                33308  1151858   679790   83767
## 6   377510      221199    61324                28928  1262516   749125   87276
## 7   375364      240933    62659                33747  1289224   757468   83640
## 8   382920      235047    65475                64439  1285788   747971   92885
##     Idaho Illinois Indiana    Iowa  Kansas Kentucky Louisiana   Maine Maryland
## 3 1839106 12812508 6785528 3190369 2937880  4505836   4657757 1362359  6177224
## 4  114128   705616  408828  190064  179446   264254    281257   61477   345047
## 5  129335   763671  442186  205867  198074   282850    300637   68948   377185
## 6  138314   831877  465339  217636  208847   298729    319364   75490   400968
## 7  134108   855690  472267  219549  208798   296815    315433   79872   400253
## 8  127524   842565  459058  216654  202972   291637    304262   74720   396606
##   Massachusetts Michigan Minnesota Mississippi Missouri Montana Nebraska
## 3       7029917 10077331   5706494     2961279  6154913 1084225  1961504
## 4        340020   548875    340357      171647   355024   59224   126605
## 5        369428   594605    366937      182950   381173   66297   135906
## 6        403461   628558    383116      207422   405032   69137   140325
## 7        449631   667560    374919      213084   407688   66565   138062
## 8        513628   674096    357908      198404   410729   66952   130698
```

```
##    Nevada New.Hampshire New.Jersey New.York North.Carolina North.Dakota
## 3 3104614       1377529    9288994 20201249       10439388       779094
## 4  174032         61480     502046  1060610         574468        52366
## 5  192429         69472     547708  1128030         625115        52879
## 6  205184         77118     592713  1191583         679932        50284
## 7  195339         84632     604078  1276802         700316        50616
## 8  193741         86990     583873  1391913         701089        59933
##       Ohio Oklahoma  Oregon Pennsylvania Rhode.Island South.Carolina
## 3 11799448  3959353 4237256     13002700      1097379        5118425
## 4   666434   241242  215252       667816        51903         277144
## 5   713483   264708  242852       731907        56266         304900
## 6   756261   280004  257649       778376        61948         329491
## 7   777343   275687  255559       822939        77125         336539
## 8   780712   270664  265951       856692        78805         327031
##   South.Dakota Tennessee    Utah Vermont Virginia Washington West.Virginia
## 3       886667   6910840 3271616  643077  8631393    7705281       1793716
## 4        57742    393767  239780   28555   481405     437231         89207
## 5        61449    419749  264449   32747   520554     473102         99929
## 6        62883    447464  280003   35271   551874     488435        107246
## 7        59482    452245  271143   40937   580169     473975        111726
## 8        57922    457172  273499   41658   591822     493488        109191
##   Wisconsin Wyoming
## 3   5893718  576851
## 4    322285   33955
## 5    352714   38016
## 6    380649   40646
## 7    386233   38415
## 8    379451   35258
```

```r
#Data Set 2:  CDC COVID Hospitalizations by Jurisdiction
URL2 <- "US_COVID-19_Hospitalization_Metrics_by_Jurisdiction.csv"
DS2 <- read.csv(URL2, header=T, stringsAsFactors=T)
DS2.1 <- select(DS2, -contains("7.Day"))
DS2.2 <- select(DS2.1, -contains("prior.week"))
DS2.3 <- select(DS2.2, -contains("Rate"))
DS2.4 <- select(DS2.3, -contains("Cumulative"))
DS2.5 <- DS2.4 %>% group_by(Collection.Date) %>%  summarise(New.COVID.19.Hospital.Admissions = sum(New.C
head(DS2.5)
```

```
## # A tibble: 6 x 3
##   Collection.Date New.COVID.19.Hospital.Admissions Total.Hospitalized.COVID.19~1
##   <fct>                                      <dbl>                         <dbl>
## 1 2020-08-01                                 16224                            NA
## 2 2020-08-02                                 12372                            NA
## 3 2020-08-03                                 12987                            NA
## 4 2020-08-04                                 14529                        117825
## 5 2020-08-05                                 14025                        118713
## 6 2020-08-06                                 14067                        116604
## # i abbreviated name: 1: Total.Hospitalized.COVID.19.Patients
```

```r
#Data Set 3:  COVID Cumulative Deaths by state and gender and age
URL3 <-  "Provisional_COVID-19_Deaths_by_Sex_and_Age.csv"
DS3 <- read.csv(URL3, header=T, stringsAsFactors=T)
```

```r
DS3.1 <- select(DS3, -contains("Pneumonia"))
DS3.2 <- select(DS3.1, -contains("Influenza"))
DS3.3 <- select(DS3.2, -contains("Footnote"))
DS3.4 <- DS3.3[-c(1:13771), ] #reduces to just data by month
DS3.5 <- select(DS3.4, -contains("Date"))
DS3.6 <- select(DS3.5, -contains("Group",))
DS3.7 <- select(DS3.6, -contains("Data"))
head(DS3.7)
```

```
##       Year Month         State       Sex COVID.19.Deaths Total.Deaths
## 13772 2020     1 United States All Sexes               0         1784
## 13773 2020     1 United States All Sexes               0         2966
## 13774 2020     1 United States All Sexes               0          315
## 13775 2020     1 United States All Sexes               0          471
## 13776 2020     1 United States All Sexes               0         2596
## 13777 2020     1 United States All Sexes               0         4426
```

```r
#Data Set 4:  COVID cumulative cases by week by state
URL4 <-  "Weekly_US_COVID-19_Cases_and_Deaths_by_State.csv"
DS4 <- read.csv(URL4, header=T, stringsAsFactors=T)
DS4.1 <- select(DS4, -contains("tot"))
DS4.2 <- select(DS4.1, -contains("historic"))
head(DS4.2)
```

```
##   date_updated state start_date   end_date new_cases new_deaths
## 1   01/23/2020    AK 01/16/2020 01/22/2020         0          0
## 2   01/30/2020    AK 01/23/2020 01/29/2020         0          0
## 3   02/06/2020    AK 01/30/2020 02/05/2020         0          0
## 4   02/13/2020    AK 02/06/2020 02/12/2020         0          0
## 5   02/20/2020    AK 02/13/2020 02/19/2020         0          0
## 6   02/27/2020    AK 02/20/2020 02/26/2020         0          0
```

```r
#Data Set 5:  COVID Case level by case, incl month, age, gender, hosp, death
ULR5 <- "COVID-19_Case_Surveillance_Public_Use_Data_with_Geography.csv"
DS5 <- read.csv(ULR5, header=T, stringsAsFactors=T)
DS5.1 <- select(DS5, -contains("fips"))
DS5.2 <- select(DS5.1, -contains("county"))
DS5.3 <- select(DS5.2, -contains("interval"))
DS5.4 <- select(DS5.3, -contains("process"))
DS5.5 <- select(DS5.4, -contains("exposure"))
DS5.6 <- select(DS5.5, -contains("symptom"))
DS5_7 <- select(DS5.6, -contains("underlying"))
DS5_7$RecCount <- 1
head(DS5_7)
```

```
##   case_month res_state      age_group    sex  race           ethnicity
## 1    2021-12        NY 50 to 64 years Female White Non-Hispanic/Latino
## 2    2022-03        MS   0 - 17 years   <NA>  <NA>                <NA>
## 3    2020-11        TX 18 to 49 years   <NA>  <NA>                <NA>
## 4    2023-11        TX 18 to 49 years   <NA>  <NA>                <NA>
## 5    2022-10        TX 50 to 64 years   <NA>  <NA>                <NA>
## 6    2022-09        SD 18 to 49 years Female  <NA>                <NA>
```

```
##              current_status hosp_yn  icu_yn death_yn RecCount
## 1 Laboratory-confirmed case      No Missing       No        1
## 2             Probable Case Missing Missing  Missing        1
## 3             Probable Case Missing Missing  Missing        1
## 4             Probable Case Missing Missing  Missing        1
## 5             Probable Case Unknown Missing  Missing        1
## 6 Laboratory-confirmed case Unknown Missing     <NA>        1
```

```
#Data Set 6:  Vaccination statuses by date and age
URL6 <- "COVID-19_Vaccinations_in_the_United_States_County.csv"
DS6 <- read.csv(URL6, header=T)
DS6.1 <- select(DS6, -contains("fips"))
DS6.2 <- select(DS6.1, -contains("county"))
DS6.3 <- select(DS6.2, -contains("booster"))
DS6.4 <- select(DS6.3, -contains("pct"))
DS6.5 <- select(DS6.4, -contains("SVI"))
DS6.6 <- select(DS6.5, -contains("census"))
DS6.7 <- select(DS6.6, -contains("metro"))
DS6.8 <- select(DS6.7, -contains("MMWR"))
head(DS6.8)
```

```
##         Date Recip_State Administered_Dose1_Recip
## 1 05/10/2023          WI                    11123
## 2 05/10/2023          IA                     3149
## 3 05/10/2023          NY                  1391226
## 4 05/10/2023          TX                    11678
## 5 05/10/2023          MI                   104075
## 6 05/10/2023          GA                     1573
##   Administered_Dose1_Recip_5Plus Administered_Dose1_Recip_12Plus
## 1                          11097                           10863
## 2                           3145                            3079
## 3                        1384503                         1329779
## 4                          11660                           11468
## 5                             NA                              NA
## 6                           1572                            1537
##   Administered_Dose1_Recip_18Plus Administered_Dose1_Recip_65Plus
## 1                           10368                            4749
## 2                            2966                            1249
## 3                         1232671                          313054
## 4                           10849                            3920
## 5                              NA                              NA
## 6                            1458                             366
##   Series_Complete_Yes Series_Complete_5Plus Series_Complete_5to17
## 1               10325                 10311                   648
## 2                2951                  2949                   157
## 3             1179481               1174924                133132
## 4               10007                  9999                   669
## 5              103137                103054                  7285
## 6                1274                  1273                    91
##   Series_Complete_12Plus Series_Complete_18Plus Series_Complete_65Plus
## 1                  10105                   9663                   4483
## 2                   2887                   2792                   1211
## 3                1127179                1041792                 268167
## 4                   9845                   9330                   3448
```

```
## 5                    101354                  95769                  32377
## 6                      1245                   1182                    319
```

What information is not self-evident?

The assumption is all data provided by the states and facilities that the federal entities have been complied is accurate. If possible validation between individual state data reporting and the federal aggregation should be completed.

What are different ways you could look at this data?

There are so many ways to look at this data. The first would be by geography - understanding which states had peaks of cases and when. From there demographics would be analyzed - is there specific age, gender, or races that were more susceptible to contract C19. From there those could both be repeated on individuals hospitalized and those who passed away from C19. There are multiple ways the data could be matrixed by these splits to look for correlation.

How do you plan to slice and dice the data?

First the data would need to be joined by like data elements to make modeling easier. Summary variables such as totals and percent of incidence (like hospitalization rate and death rate) would be very beneficial to graph prior to modeling to look for trends in the data. Also looking at the vaccination rates between one dose and completing the series would be interesting to see if that impacts the transmission, hospitalization, or death rates.

How could you summarize your data to answer key questions?

I found that it is best to review your full analysis with a peer to make sure that there is nothing missed and all logic makes sense. Feedback from them can be very beneficial. Then I would create a power point presentation to provide a summary of the data analyzed, then the key charts and model results. While this process is very time consuming, the executives what to know the impact to them and their citizens so they can best act on it.

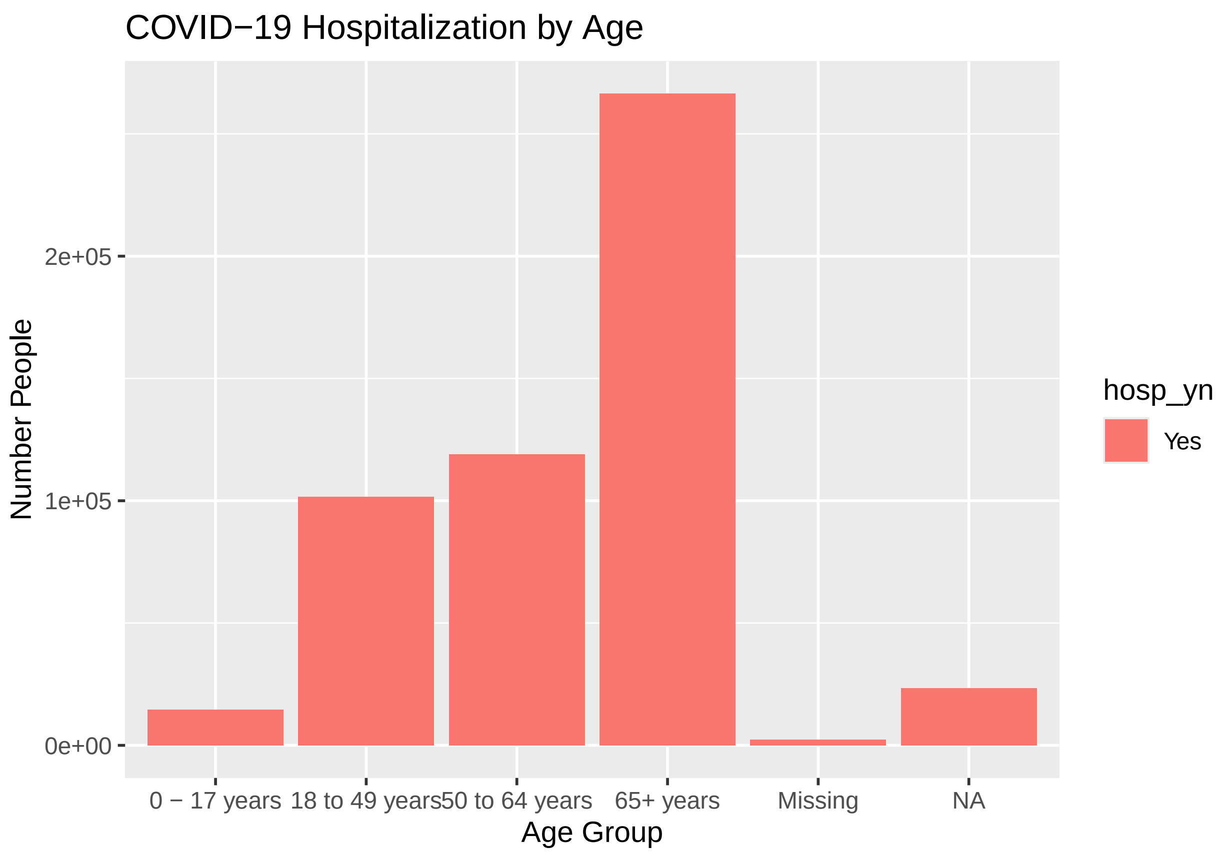
Plots and Table Needs

The types of plots that will be key in illustrating findings are bar charts, line charts, scatter plots, stacked bar charts, along with regression lines, residual plots. Correlation and covariant tables will be needed as well. Plots and tables will be generated to look at relationships between the variables. All graphs will be appropriately labeled with titles and legend, along with a text summary of what story the graphic is telling.

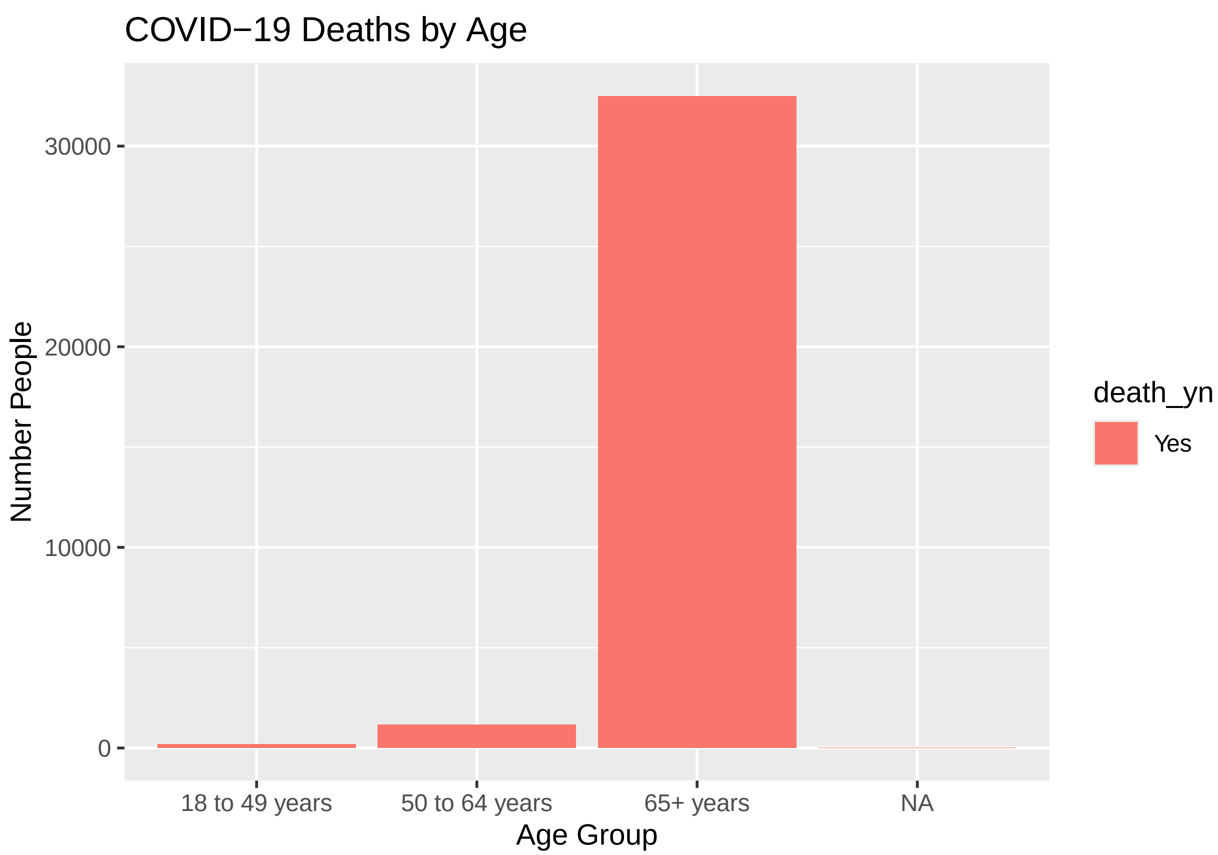
```
DS5_Summ <-  DS5_7%>%group_by(age_group,sex, race, ethnicity, hosp_yn, death_yn)%>%
  summarize(TotalCount = sum(RecCount))
```

```
## 'summarise()' has grouped output by 'age_group', 'sex', 'race', 'ethnicity',
## 'hosp_yn'. You can override using the '.groups' argument.
```

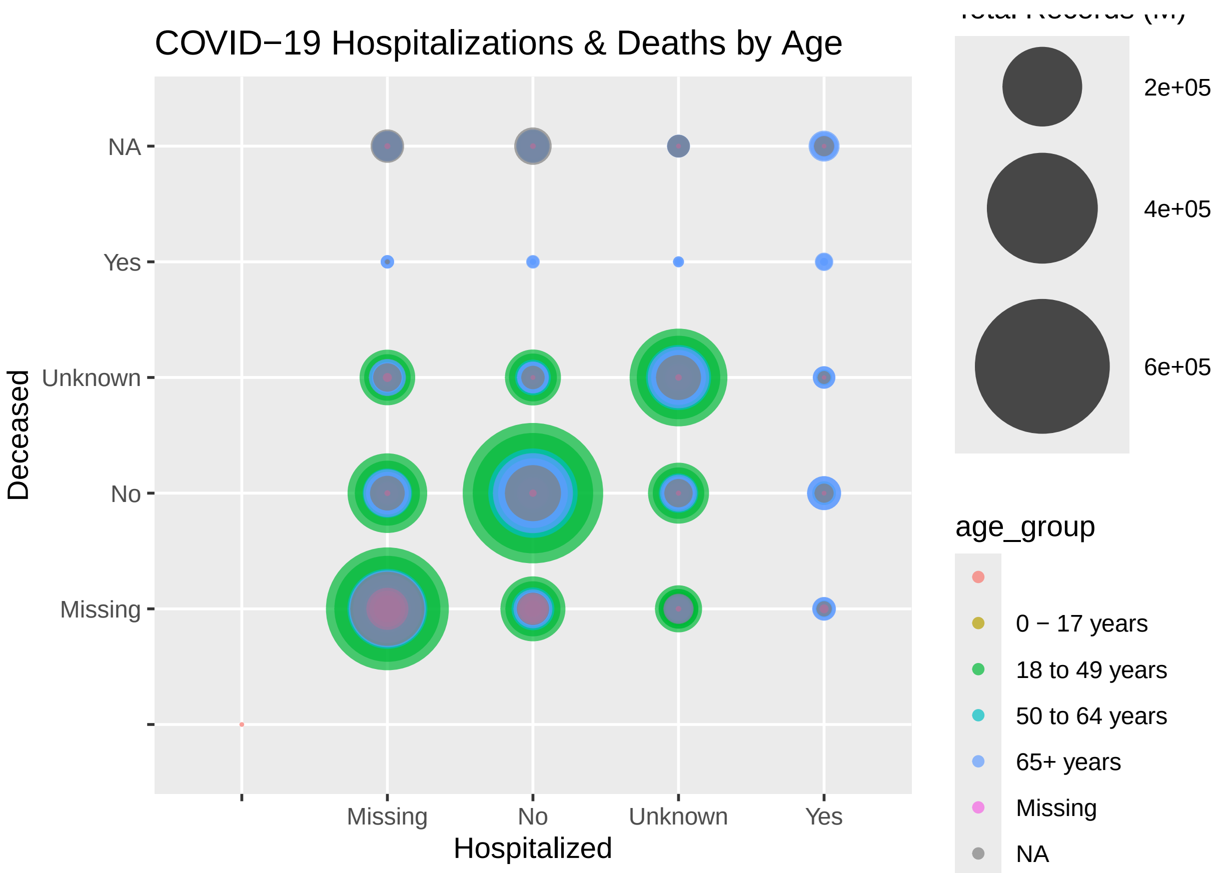
```
Hosp <- filter(DS5_Summ, hosp_yn == "Yes")
Death <- filter(DS5_Summ, death_yn == "Yes")

ggplot(Hosp, aes(fill=hosp_yn, y=TotalCount, x=age_group)) + geom_bar(position="stack", stat = "identit
  labs(x="Age Group", y="Number People",  title="COVID-19 Hospitalization by Age")
```



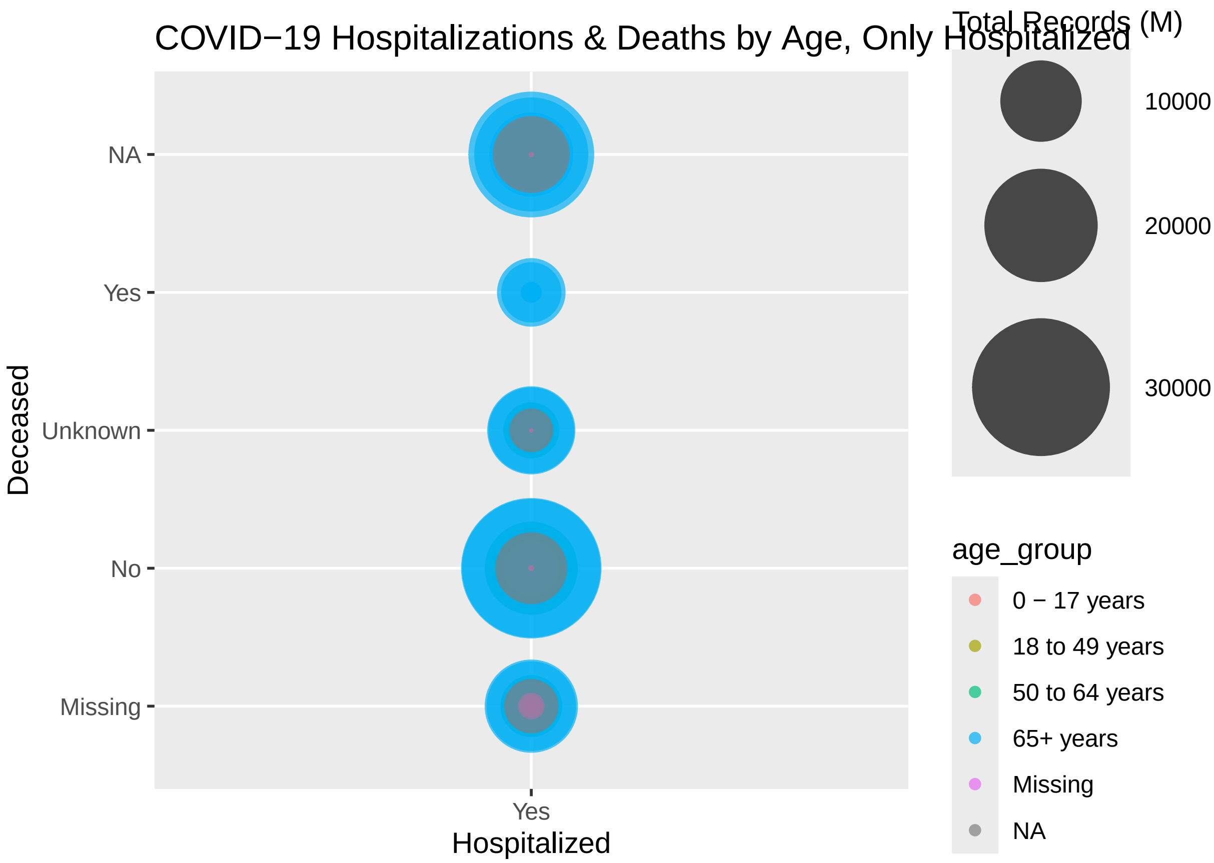

```
ggplot(Death, aes(fill=death_yn, y=TotalCount, x=age_group)) + geom_bar(position="stack", stat = "identi
  labs(x="Age Group", y="Number People",  title="COVID-19 Deaths by Age")
```




```
ggplot(DS5_Summ, aes(x=hosp_yn, y=death_yn, size = TotalCount, color = age_group)) +
    geom_point(alpha=0.7) +
    scale_size(range = c(.1, 24), name="Total Records (M)") +
  labs(x="Hospitalized", y="Deceased",  title="COVID-19 Hospitalizations & Deaths by Age")
```




```
ggplot(Hosp, aes(x=hosp_yn, y=death_yn, size = TotalCount, color = age_group)) +
    geom_point(alpha=0.7) +
    scale_size(range = c(.1, 24), name="Total Records (M)") +
  labs(x="Hospitalized", y="Deceased",  title="COVID-19 Hospitalizations & Deaths by Age, Only Hospitali
```



Do you plan on incorporating any machine learning techniques to answer your research questions? Explain.

Yes linear and logistic regression would be great tools to create a model that provides the coefficients for future pandemic response.

Questions for future steps.

- How complete is the data the federal government obtained?
- Are there variables that are to correlated that need to be evaluated?
- Is there other data available that would enhance the analysis?
- What is the easiest way to aggregate the data to the same date time frame?
- Find the data sets that provide the most value as there is a lot of overlap between the currently identified sets.
- Should the modeling should be done at the state level or country?

Analysis Insights

Those who contracted C19 who were 65 years and older had a higher concentration of hospitalizations and death. Age plays a significant factor in determining the severity of the disease.

Implications

With any medical crisis such as a pandemic, one has to be very careful when doing analaysis and creating models. They much be as accurate as possible because any prediction can hit social media and blow up in a good or bad way. If the model is not fit well it could cause panic and economic impacts through riots or rushes for health care assistance that is not available or needed.

A model must be fine tuned, tested, and evaluated on multiple data sets to ensure accuracy, then communicated out appropriately so that facilities are ready to support anyone that needs medical care.

Limitations

The greatest limitation is the type of data available to create a model. Having been part of the response efforts in Nebraska, I know that the CDC can choose how they want to report the data. A lot of the data on testing was only on specific test types and excluded a large portion of rapid test done in medical offices or homes that would make the data more robust.

Doing a public records request from each state to specify the types of test, time frame and specific demographics needed would ensure the data is representative of the entire population and pulled in the same manner. Asking for de-identified data would also help as the states would then be less at risk of a HIPAA incident.

Conclusion

The amount of C19 data available to create models and analyze is vast. Finding the correct dataset that provides the elements needed along with an accurate representation of the population is the most critical. From there, a data scientist could spend hours, days, and months evaluating relationships between the multiple variables. There is much to be learned from the concerted effort that all had in making sure data was a part of communication in the pandemic response.